

Cedar Site Bai

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★ SUMMARY

Ph.D. cand. in CS with a strong background in machine learning, optimization, and statistics. Experienced in their applications to language model pre-training, fine-tuning, recommendation, and AI safety, with **oral presentations at ICML, ICLR, COLT** and **experience at Amazon, TikTok**. Seeking a **Researcher/Scientist** role.

🧰 INDUSTRY EXPERIENCE

Amazon, Prime Video Personalization

Applied Scientist II

May. 2026 – Present

Work on **homepage recommendation**

Managers: Drs. [Yan Fu](#), [Qilin Qi](#)

- Applied Bayesian optimization to automatically tune utility-function weights for homepage recommendations
- Developed constrained optimization for reranking under diversity and share-of-voice business constraints

Applied Scientist Intern

May. 2025 – Sep. 2025

Work on **conversational recommendation**

Mentors: Drs. [Duanshun Li](#), [Zhenyu Liao](#), [Sheikh Sarwar](#)

- Designed a reward quantifying information gain via uncertainty reduction in multi-turn conversations
- Built a data-generation pipeline integrating LLM recommenders, judges, and user simulators using real-world data
- Fine-tuned LLMs with SFT, DPO, PPO, increasing conversation efficiency by 7% and hit rate by 2%

TikTok, Shop Mall Ads, Machine Learning Engineer Intern

May. 2024 – Aug. 2024

Work on **ads recommendation ranking models**

Mentor: [Du Zhang](#), Manager: [Dr. Yan Yan](#)

- Delivered two launches that improved advertiser value by 1.63% and 5.94% (**\$X00,000 in profits**) after A/B test
- Applied neural architecture search to select features/emb. sizes, reducing the model size & improving accuracy
- Adapted Deep Cross Network for improved conversion rate (CVR) prediction on sequential data

🎓 EDUCATION

• Purdue University

Jan. 2021 – May. 2026

Ph.D. in Computer Science (Machine Learning & Optimization)

Advisor: [Prof. Brian Bullins](#)

M.S. in Statistics & Computer Science (along the way)

West Lafayette, IN, US

• Xi'an Jiaotong University (XJTU)

Sep. 2016 – Jun. 2020

B.S. in Computer Science (Qian Xuesen Honors College)

Xi'an, China

Selected Awards: "Siyuan" Scholarship (twice); National 2nd Prize in Mathematical Contest in Modeling (CUMCM, top 3%)

📖 SELECTED PUBLICATION

- [1] [ICML'25, **Oral (Top 1%)**] **C. S. Bai***, A. Y. Zheng*, B. Bullins, R. A. Yeh. Model Immunization from a Condition Number Perspective. [\[PDF\]](#)
- [2] [ICLR'25, **Oral (Top 1.7%)**] **C. S. Bai**, B. Bullins. Tight Lower Bounds under Asymmetric High-Order Hölder Smoothness and Uniform Convexity. [\[PDF\]](#)
- [3] [COLT'25] **C. S. Bai**, B. Bullins. Faster Acceleration for Steepest Descent. [\[PDF\]](#)
- [4] [CIKM'26] **C. S. Bai**, D. Li, Z. Liao, S. Sarwar, H. Chen, Y. Chen, C. Yuan, H. Zhang, Q. Qi. Ask to Be Sure: Informative Interactions for Confident Multi-Turn LLM Recommendation. [\[PDF\]](#)
- [5] [ICML'25] **C. S. Bai***, X. Luo*, B. Li*, P. Drineas, R. Zhang, B. Bullins. Stacey: Promoting Stochastic Steepest Descent via Accelerated ℓ_p -Smooth Nonconvex Optimization. [\[PDF\]](#)
- [6] [IJCAI'21] H. Zhang, **S. Bai**, X. Lan, D. Hsu, N. Zheng. Hindsight Trust Region Policy Optimization. [\[PDF\]](#)

 PROFESSIONAL TECHNIQUES

Machine learning (ML): Optimizer design/analysis, reinforcement learning/RLHF, recommendation, unlearning;
Programming: Python, MATLAB (Proficient); Shell script, C/C++, R, Java, SQL, HTML;
Packages / Tools: PyTorch, NumPy, Linux Command, Git (**Proficient**); AWS, scikit-learn, TensorFlow, Pandas, Slurm.

 ADDITIONAL PUBLICATION

[1] **[Preprint]** C. S. Bai, A. Y. Zheng, R. A. Yeh, B. Bullins. Spectral Saliency for Machine Unlearning. [\[PDF\]](#)

[2] **[Preprint]** Z. Song, C. S. Bai, Z. Zhang, B. Bullins, D. Gleich. Can Entry-Wise Clipping Give Spectral Control of Stochastic Gradients?. [\[PDF\]](#)

[3] **[Preprint]** H. Zhang, T. Hu, C. S. Bai, A. Xiao, D. Hsu. Uncertainty Estimation for Visual Grounding in the Open World.

[4] **[ICML’26]** Z. Song, C. S. Bai, Z. Zhang, B. Bullins, D. Gleich. Decoupling Variance and Scale-Invariant Updates in Adaptive Gradient Descent for Unified Vector and Matrix Optimization. [\[PDF\]](#)

[5] **[ICLR’24]** S. Bai, B. Bullins. Local Composite Saddle Point Optimization. [\[PDF\]](#)

[6] **[TMLR’24]** S. Bai, C. Ke, J. Honorio. On the Dual Problem of Convexified Convolutional Neural Networks. [\[PDF\]](#)

[7] **[IROS’19]** H. Zhang, X. Lan, S. Bai, X. Zhou, Z. Tian, N. Zheng. ROI-based Robotic Grasp Detection for Object Overlapping Scenes. [\[PDF\]](#)

[8] **[IROS’19]** H. Zhang, X. Lan, S. Bai, L. Wan, C. Yang, N. Zheng. A Multi-task Convolutional Neural Network for Autonomous Robotic Grasping in Object Stacking Scenes. [\[PDF\]](#)

 SELECTED COURSES

Machine Learning (ML): Statistical ML, ML Theory, Computer Vision, Data Mining, Artificial Intelligence
Math & Stats: Optimization, Probability, Statistical Inference, Computational Stats, Multivariate Analysis
Computer Science: Data Structures, Algorithms, Database, Machine Structure, Operating Systems, Networks

 TEACHING & ACADEMIC SERVICE

Purdue CS571 Artificial Intelligence	TA (Fall 23)
Purdue CS182 Foundations of Computer Science (Discrete Mathematics)	TA (Spring 23)
Purdue CS251 Data Structures and Algorithms	TA (Fall 21, Spring 22, Fall 22)

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• ICLR 2027, 2026, 2025, 2024	Reviewer	• AAAI 2026, 2025	Reviewer
• NeurIPS 2026, 2025, 2024, 2023	Reviewer	• TPAMI	Reviewer
• ICML 2026, 2025, 2024	Reviewer	• JMLR	Reviewer
• AISTATS 2025, 2023	Reviewer	• TMLR	Reviewer